

Kubernetes Service Not Reachable SOP

Objective

Restore access to a Kubernetes service when the application is not reachable internally or externally due to service, selector, port, or ingress configuration issues.

Scope

Applicable to Kubernetes workloads, platform operations, DevOps teams, SRE teams, application teams, and synthetic incident workflows across development, staging, and production-like environments.

Pre-requisites

- 1. Access to Kubernetes cluster using kubectl and the correct namespace context.
- 2. Access to application, pod, deployment, service, ingress, and node logs where applicable.
- 3. Access to monitoring dashboards such as Grafana, Prometheus, cloud monitoring, or container insights.
- 4. Permission to inspect manifests, secrets, configmaps, RBAC bindings, PVCs, and rollout history.
- 5. Escalation contact for DevOps, platform, infrastructure, network, and application teams.

Incident Metadata

Field	Value
Ticket ID	T006
Issue	Kubernetes service not reachable
Category	Networking
Service	Kubernetes
Severity	High
Status	Resolved
Description	Service not accessible externally
Expected Resolution	Check service type and ingress configuration

Procedure

- 1. Confirm whether the service is unreachable from inside the cluster, outside the cluster, or both.
- 2. Check service type, cluster IP, external IP, ports, targetPort, and selectors.
- 3. Verify that service selectors match labels on running and ready pods.
- 4. Test connectivity from an in-cluster debug pod using curl or nc.
- 5. Check endpoint objects to confirm pod IPs are attached to the service.
- 6. Review ingress, load balancer, firewall, and DNS configuration for external access issues.
- 7. Correct service selector, port mapping, ingress, or load balancer configuration.
- 8. Retest connectivity and monitor request success rate.

Common Commands

```
kubectl config current-context
kubectl get pods -n <namespace> -o wide
kubectl describe pod <pod-name> -n <namespace>
kubectl logs <pod-name> -n <namespace> --tail=100
kubectl get events -n <namespace> --sort-by=.lastTimestamp
kubectl get deploy,svc,ingress,pvc -n <namespace>
```

Troubleshooting Matrix

Issue	Possible Cause	Resolution
Service unreachable	Selector does not match pod labels	Update selector or pod labels
No endpoints	Pods not ready or label mismatch	Fix pods and validate endpoints
Wrong port	targetPort mismatch	Correct service port mapping
External access fails	Ingress or load balancer issue	Check ingress rules and external IP

Best Practices

1. Keep service selectors stable and documented.
2. Use named ports consistently.
3. Test service access from inside the cluster.
4. Monitor endpoint count and HTTP availability.

References

[Kubernetes Docs](#) | [Internal Tickets CSV](#) | [Cluster Events](#) | [Monitoring Dashboards](#) | [Application Logs](#) | [Platform Runbook](#)